

Samundra Karki

Scientific Machine Learning · Computational Mechanics · Uncertainty Quantification

Ames, IA | samundra@iastate.edu | [Website](#) | [GitHub](#) | [Google Scholar](#) | [ORCID](#)

Ph.D. candidate connecting learned geometry with reliable physics simulation. Research spans simulation directly on neural signed-distance fields, a priori error analysis for neural geometry in PDEs, closed-form neural-geometry editing, geometry uncertainty propagation, and multi-physics inverse design.

Education

Ph.D., Mechanical Engineering, Iowa State University Expected Fall 2026
GPA: 4.0/4.0 | Advisor: Prof. Baskar Ganapathysubramanian

B.E., Mechanical Engineering, Tribhuvan University May 2022
Institute of Engineering, Pulchowk Campus, Nepal

University Gold Medal: top graduate across all engineering campuses.

Peer-Reviewed Publications

- [1] **S. Karki**, M.-C. Hsu, A. Krishnamurthy, and B. Ganapathysubramanian. “Mechanics simulation with Implicit Neural Representations of complex geometries.” *Computer-Aided Design*, **190**:103978, 2026. doi:10.1016/j.cad.2025.103978.
- [2] **S. Karki**, M. Shadkhah, C.-H. Yang, A. Balu, G. Scovazzi, A. Krishnamurthy, and B. Ganapathysubramanian. “Direct flow simulations with implicit neural representation of complex geometry.” *Computer Methods in Applied Mechanics and Engineering*, **446**:118248, 2025. doi:10.1016/j.cma.2025.118248.
- [3] R. Tali, A. Rabeh, C.-H. Yang, M. Shadkhah, **S. Karki**, A. Upadhyaya, S. Dhakshinamoorthy, M. Saadati, S. Sarkar, A. Krishnamurthy, C. Hegde, A. Balu, and B. Ganapathysubramanian. “FlowBench: A Large Scale Benchmark for Flow Simulation over Complex Geometries.” *Data-centric Machine Learning Research*, **2**(14):1–35, 2025. [Publisher full text](#).
- [4] M. Shadkhah, C.-H. Yang, **S. Karki**, and B. Ganapathysubramanian. “Octree-Based Shifted Boundary Method: Evaluating the impact of hanging-node removal on convergence and solver performance for linear PDEs.” *Advances in Computational Science and Engineering*, **4**:119–141, 2025. doi:10.3934/acse.2025014.
- [5] **S. Karki**, S. Satyal, K. P. Rijal, P. Koirala, and N. Adhikari. “Comparative CFD analysis of Kali-Gandaki ‘A’ Francis runner with runner generated from Bovet method.” *IOP Conference Series: Earth and Environmental Science*, **1037**(1):012007, 2022 (3rd IAHR-Asia Symposium on Hydraulic Machinery and Systems). doi:10.1088/1755-1315/1037/1/012007.
- [6] **S. Karki**, A. Krishnamurthy, and B. Ganapathysubramanian. “Neural Geometry for PDEs: Regularity, Stability, and Convergence Guarantees.” *ICLR 2026 Workshop on AI and Partial Differential Equations (AI&PDE)*, poster, 2026. arXiv:2602.02271.

Submitted Manuscripts and Preprints

- [7] **S. Karki**, A. Krishnamurthy, and B. Ganapathysubramanian. “Propagating Geometry Uncertainty from a Single Implicit Neural Representation through a Differentiable PDE Solver.” *Submitted to ICLR 2027*.
- [8] **S. Karki**, A. Krishnamurthy, and B. Ganapathysubramanian. “GENIE: Gram-Eigenmode INR Editing with Closed-Form Geometry Updates.” *Preprint*, 2026. arXiv:2603.29860.

Manuscripts in Preparation

- [9] **S. Karki**, A. Biswas, N. Masud, M. H. H. Hasib, A. Sarkar, and B. Ganapathysubramanian. “From Sparse AFM Observations to Probabilistic Macroscale Mechanics: A Generative-Physics Framework for Plant Cell Walls.” *In preparation*.
- [10] **S. Karki**. “Suppression Is Not Sufficiency: Separating Behavioral Control from Reconstruction in Fine-Tuned Transformers.” *In preparation*.

Research and Engineering Experience

Graduate Research Assistant

Aug. 2023 – Present

Iowa State University | Advisor: Prof. Baskar Ganapathysubramanian

- Developed shifted-boundary finite-element simulation on adaptive octree meshes that queries neural SDFs directly, without surface-mesh generation, for incompressible Navier–Stokes and linear elasticity [1,2].
- Validated 2D flow against body-fitted solutions for $Re = 100\text{--}5000$ and 3D flow against triangle-mesh SBM; showed second-order L^2 convergence for elasticity and scaled to ~ 3 million DOFs [1,2].
- Derived a priori error estimates linking uniform INR error to elliptic PDE accuracy, implying INR tolerance $\varepsilon \sim h^{k+1}$ to preserve FEM convergence [6].
- Co-authored FlowBench, 10,000+ fully resolved 2D/3D flow and heat-transfer simulations over complex geometries for benchmarking neural simulators [3].

Research Scientist Intern

May 2026 – Aug. 2026

Argonne National Laboratory | Advisor: Dr. Gary Hu

- Built a modular diffusion framework for multi-physics inverse design, combining joint objective conditioning with differentiable forward surrogates.
- Combined classifier-free guidance, diffusion posterior sampling, and conflict-free gradient composition; analyzed per-objective gradient dynamics to assess feasibility during sampling.

AI Research Fellow

Jan. 2023 – Aug. 2023

FuseMachine AI, Kathmandu, Nepal

- Developed predictive-maintenance pipelines using classical ML and deployed inference as FastAPI microservices.

Computational Systems Engineer (Founding Team)

Nov. 2021 – Aug. 2023

Zebec & Mokshya, Remote

- Built Rust and TypeScript backends for financial computation, emphasizing numerical correctness, memory safety, and reliable services.

Technical Skills

Languages: C++17/20, Python, CUDA, Rust, TypeScript/JavaScript

Scientific computing: FEM/FVM, shifted boundary methods, octree AMR, PETSc, MPI, OpenMP

Machine learning: PyTorch, JAX, LibTorch, ONNX Runtime, diffusion models, neural SDFs

Teaching

Instructor, MLOps (non-credit professional course), Translational AI Center, Iowa State University

Fall 2024

Selected Conference Presentations

USNCCM18, Chicago, IL, July 2025: Advancing mechanics simulation with AI-based geometric representation.

MLCAS 2025, Tokyo, Aug. 2025: Mechanics-informed phenotyping of maize stalks using octree-based elastic simulations.

WCCM-PANACM 2024, Vancouver, July 2024: Direct simulation of Navier–Stokes on neural SDFs using the shifted boundary method.